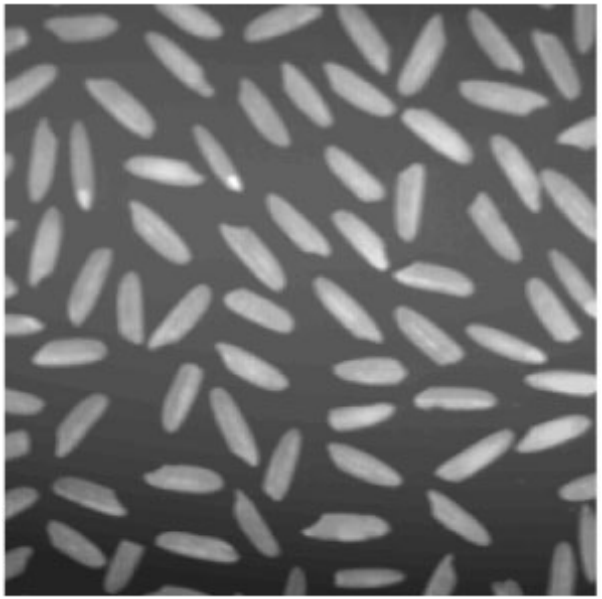
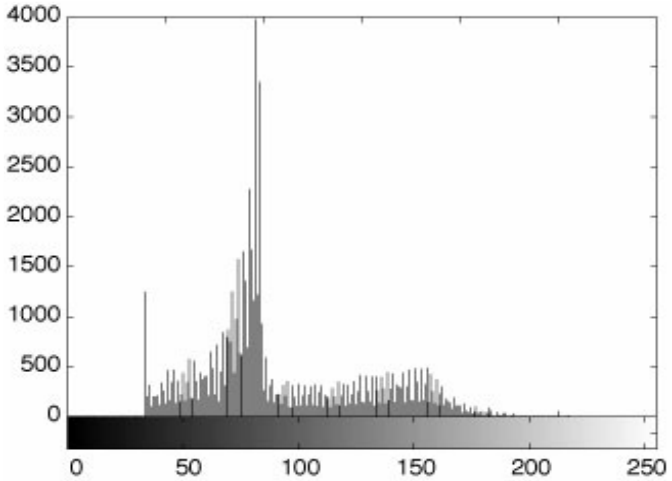
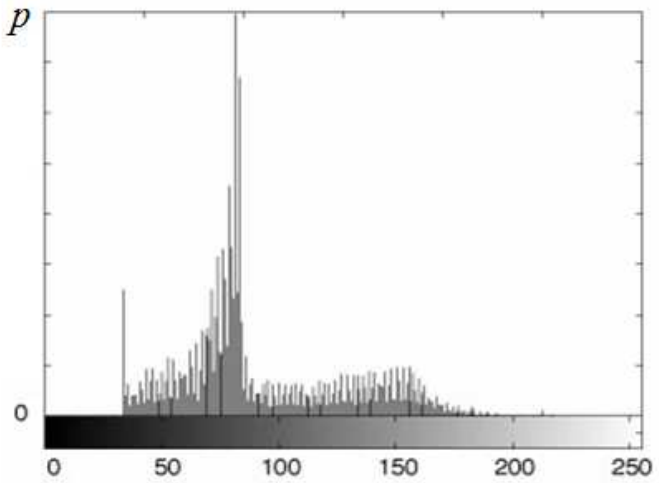
◦ Otsu's thresholding method





Rice image : `rice.tif`

Describe the histogram as a probability distribution by $p_i = n_i / N$



Let t be the determined threshold value (t unknown so far).

Define $m = \sum_{i=0}^{L-1} i \cdot p_i, \quad m_a(t) = \sum_{i=0}^{t} i \cdot p_i$

$$a(t) = \sum_{i=0}^{t} p_i, \quad b(t) = \sum_{i=t+1}^{L-1} p_i$$

Find t such that

$$\max_t \{ \frac{(m_a(t) - ma(t))^2}{a(t)b(t)} \}$$

